Assignment Title: Multi-Tenant Manufacturing Inventory & Order Fulfillment Engine
Context: Pactle serves manufacturers. A manufacturer may sell finished goods, consume raw materials, manage stock across warehouses, and fulfill customer orders. The system must prevent overselling, calculate product cost, reserve inventory, and maintain a reliable stock ledger.

Candidate Brief

Build a simplified multi-tenant inventory and order fulfillment engine for a manufacturing ERP.

The system should allow a tenant to:

1. Manage products, raw materials, and finished goods.
2. Define a bill of materials.
3. Track stock across warehouses.
4. Create customer sales orders.
5. Reserve stock for orders.
6. Manufacture finished goods from raw materials.
7. Maintain an auditable stock ledger.

Core Requirements

1. Multi-Tenant Support

Every entity must belong to a tenant:

- Products
- Warehouses
- Inventory balances
- Stock ledger entries
- Bill of materials
- Customers
- Sales orders
- Manufacturing orders

Tenant A must never see or affect Tenant B's stock.

---

2. Product Catalog

Support three product types:

```
raw_material
finished_good
trading_good
```

Product fields:

```
name
sku
type
unit_of_measure
standard_cost
selling_price
active
```

Rules:

- SKU must be unique per tenant.
- Inactive products cannot be ordered or manufactured.
- Raw materials cannot be sold unless explicitly marked as sellable.
- Finished goods can have a bill of materials.

---

3. Warehouses

A tenant can have multiple warehouses.

Warehouse fields:

```
name
code
city
active
```

Rules:

- Warehouse code must be unique per tenant.
- Inactive warehouses cannot be used for new stock movements.

---

4. Stock Ledger

All inventory changes must be recorded as ledger entries.

Ledger entry fields:

```
tenant_id
warehouse_id
product_id
movement_type
quantity_change
unit_cost
reference_type
reference_id
created_at
created_by
```

Movement types:

```
purchase_receipt
sales_reservation
sales_dispatch
reservation_release
manufacturing_issue
manufacturing_receipt
stock_adjustment
```

Rules:

- Do not directly update stock without a ledger entry.
- Current stock should be derived from ledger entries or maintained with ledger-backed balances.
- Quantity cannot go negative unless explicitly documented as allowed.
- Each stock movement must be tenant-scoped.

---

5. Inventory Balance

Expose current inventory by product and warehouse.

Required fields:

```
on_hand_quantity
reserved_quantity
available_quantity
average_cost
```

Formula:

```
available_quantity = on_hand_quantity - reserved_quantity
```

The candidate may either:

- calculate from ledger on demand, or
- maintain an inventory balance table updated transactionally.

They must explain the tradeoff.

---

6. Purchase Receipt

Create an API to receive stock for raw materials or trading goods.

Example:

```
{
  "tenant_id": "tenant_123",
  "warehouse_id": "wh_1",
  "items": [
    {
      "sku": "RM-STEEL-001",
      "quantity": 100,
      "unit_cost": 50
    }
  ]
}
```

Expected:

- Creates stock ledger entries.
- Increases on-hand stock.
- Updates average cost.

---

7. Bill of Materials

Allow tenants to define BOM for finished goods.

Example:

```
Finished Good: CNC Machine
BOM:
- Steel Plate: 5 kg
- Motor: 1 unit
- Control Panel: 1 unit
```

BOM fields:

```
finished_good_id
raw_material_id
quantity_required
wastage_percent
```

Rules:

- BOM can only be defined for finished goods.
- BOM components must be active products.
- Quantity required must be greater than zero.
- Prevent circular BOM references if nested BOM is supported.

---

8. Manufacturing Order

Create a manufacturing order for a finished good.

Example:

```
{
  "tenant_id": "tenant_123",
  "warehouse_id": "wh_1",
  "finished_good_sku": "FG-CNC-001",
  "quantity_to_produce": 10
}
```

When completed:

- Check raw material availability based on BOM.
- Issue raw materials from inventory.
- Receive finished goods into inventory.
- Create ledger entries:
    - `manufacturing_issue` for raw materials
    - `manufacturing_receipt` for finished goods
- Calculate finished good cost based on consumed raw material cost.
- If insufficient raw materials, reject completion with detailed shortage list.

Shortage response example:

```
{
```

```
  "can_complete": false,
  "shortages": [
    {
      "sku": "RM-STEEL-001",
      "required": 50,
      "available": 20,
      "shortage": 30
    }
  ]
}
```

---

9. Sales Order

Create a sales order for customers.

Sales order fields:

```
customer_id
order_number
status
items
subtotal
tax
grand_total
```

Statuses:

```
draft
confirmed
partially_dispatched
dispatched
cancelled
```

Rules:

- Order number must be unique per tenant.
- Products must be active and sellable.
- Totals must be calculated server-side.
- Sales order cannot dispatch more than ordered quantity.

---

10. Stock Reservation

When a sales order is confirmed:

- Reserve available stock for each item.
- Create `sales_reservation` ledger entries or reservation records.
- Increase reserved quantity.
- If insufficient stock, return shortages.
- Prevent two orders from reserving the same available stock.

Important:

- This must be transaction-safe conceptually.
- Candidate should explain how they would handle concurrency.

---

11. Dispatch

When dispatching a sales order:

- Reduce on-hand stock.
- Reduce reserved stock.
- Create `sales_dispatch` ledger entry.
- Update order status.
- Allow partial dispatch.
- Do not dispatch unreserved or unavailable stock.

---

12. Cancellation

When cancelling a confirmed order:

- Release reserved stock.
- Create `reservation_release` ledger entry.
- Update order status to cancelled.
- Do not allow cancellation after full dispatch.

Required APIs

Minimum:

```
POST /products
```

```
GET /products

POST /warehouses
GET /warehouses

POST /purchase-receipts

POST /bom
GET /bom/:finished_good_id

POST /manufacturing-orders
POST /manufacturing-orders/:id/complete

POST /sales-orders
POST /sales-orders/:id/confirm
POST /sales-orders/:id/dispatch
POST /sales-orders/:id/cancel

GET /inventory
GET /stock-ledger
```

Required Seed Data

Candidate should include:

- 2 tenants
- 2 warehouses per tenant
- Raw materials:
    - Steel Plate
    - Motor
    - Control Panel
- Finished goods:
    - CNC Machine
    - Hydraulic Press
- Trading goods:
    - Safety Kit
- Customers
- Initial stock receipts
- BOM for at least one finished good

Example Test Scenario

Step 1: Receive raw materials

Receive:

```
100 kg Steel Plate
20 Motors
20 Control Panels
```

Step 2: Define BOM

CNC Machine requires:

```
5 kg Steel Plate
1 Motor
1 Control Panel
```

Step 3: Manufacture 10 CNC Machines

Expected:

- Consume 50 kg Steel Plate.
- Consume 10 Motors.
- Consume 10 Control Panels.
- Add 10 CNC Machines.
- Create all ledger entries.
- Calculate cost.

Step 4: Create sales order for 8 CNC Machines

Expected:

- Confirming order reserves 8 CNC Machines.
- Available stock becomes 2.
- Reserved stock becomes 8.

Step 5: Try creating another order for 5 CNC Machines

Expected:

- Reject or partially allow based on design.
- Must show shortage of 3.

Step 6: Dispatch first order

Expected:

- On-hand decreases.
- Reserved decreases.
- Ledger shows dispatch.
- Order status updates.

Bonus Requirements

Bonus 1: Average Cost Calculation

Maintain moving average cost after purchase receipts and manufacturing receipts.

Bonus 2: Partial Manufacturing

Allow manufacturing order to be partially completed.

Bonus 3: Inventory Valuation Report

Endpoint:

```
GET /inventory/valuation
```

Shows:

```
{
  "total_inventory_value": 250000,
  "by_product": [
    {
      "sku": "FG-CNC-001",
      "quantity": 10,
      "average_cost": 12000,
      "value": 120000
    }
  ]
}
```

Bonus 4: Low Stock Alerts

Return products where available quantity is below reorder level.

Bonus 5: Audit Trail

Track:

- Product created
- Stock received
- BOM created
- Manufacturing completed
- Sales order confirmed
- Stock reserved
- Stock dispatched
- Reservation released